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MEMORANDUM

September 4, 2015

TO: Jonathan Akins, P.E.
Ryan Heiland
University of Colorado – Facilities Management

FROM: Jason Klausner, PE, CLIA
Cullen Kinoshita, CLIA
Robert Beccard, PE

RE: **University of Colorado Newton Court and Athens Court Raw Water Irrigation System Master Plan**

DRAFT

Background and Summary

University of Colorado Newton Court and Athens Court (Netwon-Athens), located at the southwest quadrant of the Arapahoe Avenue and Folsom Street intersection, will include a future raw water irrigation system consisting of irrigation water storage, pump station, wet well, and various piping (ie. overflow, intake pipe, mainline pipe, etc.). A Site Map is provided on the next page.

An existing 10 horsepower (HP) pump (3 phase, 208 VAC) provides raw water from the Smith Goss Ditch to the Newton Court irrigation system. The existing pump and associated infrastructure (ie. wet well, connection to Smith Goss ditch) will be removed and a new raw water irrigation system will provide water to the Newton Court irrigation system. A connection will also be made to the existing irrigation mainline that supplies water to the Practice Football Fields.

A stub out will be provided from the new raw water irrigation system intended for future expansion to Athens Court. The existing irrigation systems that serve Athens Court are currently served by numerous potable water connections. As the new mainline is extended to the west, the potable water sources are intended to be disconnected and connected to the new raw water irrigation system.

Raw water from the Smith Goss ditch system can be stored in either an irrigation pond or an underground storage vessel. The storage unit (pond or vessel) could be located at numerous locations within the vicinity of the Practice Football Fields and Student Housing area and will provide a maximum three (3) days of storage. Overflow piping can drain south towards Boulder Creek or potentially north to Smith Goss Ditch piping located underneath Arapahoe Avenue. Additional investigation is required to determine if overflow can discharge to Arapahoe Avenue.

The new pump station consists of 2, 40 HP main pumps and a 3 HP submersible pressure maintenance pump all placed within an enclosure or building. State-of-the-art electrical controls will provide power and manage the pump station.

The estimated cost for raw water irrigation system including the storage unit (pond or vessel), pump station, enclosure, wet well, mainline piping, and associated equipment ranges $600,000 to $2,300,000 depending on desired storage unit type (pond or vessel) and preferred overflow pipe routing option.
The remainder of this document presents the following:

- Irrigation Water Use Analysis
- Irrigation Pond (Storage Option #1)
- Irrigation Vessel (Storage Option #2)
- Irrigation Pump Station
- Opinions of Probable Cost

Site Map



Note: Site map from Google Earth.

Irrigation Water Use Analysis

The irrigation water use analysis below provides insight to irrigation storage and pump station capacity requirements. The following assumptions were used:

- 65% irrigation system application efficiency
- 75% tap utilization efficiency
- 0.25 in/day peak evapotranspiration (ET) rate
- 5 days of allowable irrigation per week
- 8 hours per day irrigation window
- 0.8 crop coefficient for turf
- 0.7 crop coefficient for drought tolerant turf
- 0.5 crop coefficient for native grass
- 0.5 crop coefficient for plantings

University of Colorado representatives provided approximated irrigated areas for Newton-Athens that accounts for future master planned areas that included:

1. Parcels A1-B6
    - 6.32 total acres
        - 15% Turf – 0.95 acres
        - 50% Plantings – 3.16 acres
2. Creek
    - 4.5 total acres
        - 25% drought tolerant turf – 1.13 acres
        - 65% native grass – 2.93 acres
3. Park
    - 1.05 total acres
        - 40% turf – 0.42 acres
        - 10% plantings – 0.11 acres
4. Sports Fields
    - 12.8 total acres
        - 85% turf – 10.9 acres

Aqua Engineering estimated the area north of the Sports Fields at 5 acres and is assumed turf.

Therefore the total irrigated areas, by plant type, are:

- Turf – 17.25 acres
- Drought Tolerant Turf – 1.13 acres
- Plantings – 3.27 acres
- Native Seed – 2.93 acres

Table 1 illustrates the irrigation water use analysis.

Table 1: Water Use Analysis

T A B L E 1: WATER USE ANALYSIS

Aqua Engineering, Inc.
375 E. Horsetooth Rd, Bldg 2-202
Fort Collins, CO 80525-3196

July 30, 2015
Project Name: CU Newton Athens
Location: Boulder, CO
Prepared By: JHK

	Turf	Drought Tolerant Turf	Plantings	Native Grass	Totals
AREA , acres	17.25	1.13	3.27	2.93	24.56
PEAK SEASON DESIGN					
PLANT WATER REQUIREMENT, inches/day	0.20	0.18	0.13	0.13	NA
OPERATING LOSS, inches	0.11	0.09	0.07	0.07	NA
TOTAL DAILY APPLICATION REQUIREMENT, inches	0.31	0.27	0.19	0.19	NA
TOTAL DAILY APPLICATION REQUIREMENT, acre*ft	0.44	0.03	0.05	0.05	0.57
TOTAL DAILY APPLICATION REQUIREMENT, gallons	144,110	8,225	17,050	15,274	184,659
SEASONAL PLANT WATER REQUIREMENTS, inches	24.0	21.0	15.0	15.0	NA
SEASONAL EFFECTIVE PRECIPITATION, inches	7.0	7.0	7.0	7.0	NA
TOTAL SEASONAL IRRIGATION APPLICATION, inches	17.0	14.0	8.0	8.0	47.0
TOTAL SEASONAL IRRIGATION APPLICATION, acre*ft	24.4	1.3	2.2	2.0	29.9
TOTAL SEASONAL IRRIGATION APPLICATION, gallons	7,962,102	427,000	710,000	635,000	9,734,102
IRRIGATION FLOW REQUIREMENT WITH AN IRRIGATION WINDOW OF 6 HOURS, 5 DAYS A WEEK (gpm)	747	43	88	79	957
IRRIGATION FLOW REQUIREMENT WITH AN IRRIGATION WINDOW OF 8 HOURS, 5 DAYS A WEEK (gpm)	560	32	66	59	718
IRRIGATION FLOW REQUIREMENT WITH AN IRRIGATION WINDOW OF 10 HOURS, 5 DAYS A WEEK (gpm)	448	26	53	48	574

The peak irrigation demand ranges 500-1000 GPM depending on irrigation window. For the remainder of this document, a flow of 800 GPM will be used for preliminary design estimates.

Irrigation Pond (Storage Option #1)

The minimum irrigation pond storage is based on three (3) days of peak irrigation application requirement and the irrigation water use analysis presented above. Below are the inputs used to size the irrigation pond:

- Peak daily irrigation application requirement (From Table 1 above)= 0.57 acre-feet of water
- Three (3) days of storage
- 1.15 safety factor to account for storage losses (ie. intake structure, particle settlement, etc.)
- 4 (Horizontal, H):1 (Vertical, V) side slope for first 12 horizontal feet. 3:1 slope thereafter for a total pond depth of 8-feet.

Based on the above inputs the pond storage requirement and estimated footprint area are:

1. Pond Storage Requirement = 1.96 acre-ft
2. Pond Footprint Area = 0.39 acres (17,000 sf) – equivalent to a circular pond with a diameter of approximately 150 feet.

It is recommended that the irrigation pond be lined with either a 30 MIL RPE (Reinforced Polyethylene) or 30 MIL PVC liner to minimize percolation losses. An 8-ounce geotextile fabric is recommended on each side of the liner to offer protection. A fence around the pond is recommended for liability purposes and the fence must be placed 10-feet minimum from the pond edge to account for the pond liner key and maintenance access.

The irrigation pond is to be flow-through that allows water delivered to overflow to Boulder Creek or possibly back to Smith Goss Ditch underneath Arapahoe Avenue (Refer to Figure 1). If the pond is not full the supplied water is either consumed by irrigation or fills the pond. Per discussion with University of Colorado Representatives, approximately 0.3 cfs (134 GPM) is diverted from Smith Goss Ditch to Newton-Athens and flows continuously throughout the irrigation season.

It is recommended that a portion of the existing raw water delivery piping that is routed through Newton-Athens water delivery be rerouted from the existing concrete vault shown on Figure 1. The existing vault acts similarly to a manhole allowing access to the piped sections of the delivery pipeline for maintenance purposes or where there are changes in piping direction. An 8-inch HDPE DR32.5 PE4710 resin pipe is recommended to supply water from the existing concrete vault to the irrigation pond.

An overflow pipe is required from the pond to either the following locations listed below. A 12-inch HDPE DR32.5 PE4710 resin pipe is recommended to convey overflow. There are two overflow options:

1. Connect to existing concrete vault located southeast of the proposed pump station. Overflow would continue to flow through existing piping to Boulder Creek.
2. Connect to Smith Goss Ditch in Arapahoe Avenue. It is not known if the existing Ditch piping invert allows gravity flow so this would need confirmed if this option is desired.

Overflow Option 1 does not allow water discharged to Boulder Creek to be reused easily for other potential projects. Option 2 allows unused water to continue in University of Colorado's raw water delivery system for potential use.

There are various locations that can be suitable for the irrigation pond. Per discussion with University of Colorado Representatives, most of the existing Newton Court buildings are to be demolished which leaves sufficient space for the irrigation pond. The maximum elevation that can be utilized is approximately 5,306 ft so that water will gravity flow from the proposed supply connection (approximate invert elevation 5,307 ft) to the irrigation pond. Image 1 below illustrates the approximate boundary where the irrigation pond can be located factoring in available open area after Newton Court building deconstruction and existing site elevations.

Image 1: Approximate Irrigation Pond Location Boundary



Note: Image from Google Earth.

Water from the irrigation pond is conveyed through a recommended 16-inch HDPE DR32.5 PE4710 resin intake pipe to a 6-foot diameter precast concrete wet well that supplies water to the irrigation pump station. Flow velocity through the intake pipe is less than the industry standard of 2 ft/s. An oversized self flushing 30 mesh intake screen with a maximum flow rating equal to a minimum of twice the peak irrigation demand will be installed on the intake pipe end located near the pond bottom. Recommended intake screen manufacturer and model: Lakos Plum Creek PC3424 – rated at 1,700 GPM.

A pond aeration system is recommended to enhance water quality in the irrigation pond.

Irrigation Storage Vessel (Storage Option #2)

The minimum irrigation vessel storage is based on three (3) days of peak irrigation application requirement and the irrigation water use analysis presented above. Below are the inputs used to size the irrigation storage vessel:

- Peak daily irrigation application requirement (From Table 1 above)= 0.57 acre-feet of water
- Three (3) days of storage
- 1.15 safety factor to account for storage losses (ie. sloped piping, particle settlement, etc.)
- Based on the above inputs the storage requirement is 1.96 acre-ft (640,000 gallons)
- Total linear feet of 5-foot diameter piping required = 4,350 ft

The storage vessel would consist of numerous 5-foot diameter corrugated HDPE pipes manifolded together placed below grade. Water would be allowed to fill the corrugated HDPE pipes and overflow to either Boulder Creek or Smith Goss Ditch in Arapahoe Avenue. The storage vessel system can be located within the same proposed irrigation pond location shown in Image 1 above. Refer to Figure 2 for additional information.

The storage vessel requires water supply and overflow that would be provided in similar fashion as the irrigation pond discussion above. A wet well and pump station would be required within the near vicinity to the storage vessel.

A potential concern is filtering the water prior to delivery to the pump station. It is possible that the self flushing intake screen be located within a separate concrete vault. The concrete vault would be connected to the storage vessels and the wet well via separate 16-inch HDPE DR32.5 4710 resin pipes.

Water stored in the vessel must be managed so that it is consumed by irrigation within 3 days to reduce potential bacterial growth (ie. Legionella) that can be a human health liability. Water quality testing is recommended to monitor water quality for bacterial growth. It is recommended that chlorine tablets be routinely added to the storage vessel system to minimize human health liabilities.

Irrigation Pump Station

The irrigation pump station is proposed to consist of the following:

- Manufacturer: Flowtronex (Preferred manufacturer per discussion with University of Colorado)
- Type: Vertical turbine main pumps and submersible pressure maintenance pump
- Controls: Variable Frequency Drive, Programmable Logic Controller, State-of-the-Art Electronics housed within NEMA 4 enclosure located within pump enclosure
- Two (2) main 40 HP pumps and one (1) 3 HP submersible pressure maintenance pump placed on prefabricated skid
- Water level sensors within wet well
- Electromagnetic flow sensor
- Piping and valves
- Wet well hatch
- Dog leg discharge pipe
- Intake screen flushing valve, pressure regulator, and piping
- Winterization connection
- Pump station housed within an enclosure or building. The enclosure or building must include access hatch centered over wet well for removal of pumps.
- Pump enclosure or building to be sized to accommodate maintenance space requirements and aeration system equipment.
- Reinforced concrete pad supports enclosure or building and pumping equipment
- Estimated Electrical Power Requirements: 460/480 VAC, 3 Phase, 85-100 full load amps, 35-50 KVA.

Based on Table 1 results and hydraulic analysis, the pump station delivers 800 GPM at 90 PSI discharge pressure. The discharge pressure allows approximately 60 PSI minimum at sprinkler heads (ie. large radius rotors for Football Practice Fields) after considering pressure losses (ie. pipe, fittings, etc.) at the west end of the site (See Figure 3). Looped system hydraulics was used so that half the flow (400 GPM) was delivered to each side of the recommended 8-inch mainline loop that surrounds the site.

The recommended mainline pipe is DR11 HDPE PE 4710 resin that is only a distribution pipe. No remote control valves connect to the distribution pipe, only gate valves, air vacuum relief valves, or potentially drain valves are allowed on the mainline pipe. The remote control valves to be installed on Class 200 PVC submainline that connects to the distribution pipe as needed. An example of this is illustrated on Figure 3.

The estimated main pump motor horsepower requirement assumed a 65% pumping efficiency. Therefore the total horsepower requirements are:

$$HP = \frac{Flow, gpm\ X\ Head, ft}{3960\ X\ Efficiency} = \frac{800\ gpm\ X\ 90\ PSI\ X 2.31 \frac{ft}{psi}}{3960\ X\ 0.65} = 64.6\ HP$$

Since two (2) main pumps are recommended, each main pump needs to be 40 HP accounting for available motor sizes. Since the two main pumps provide a total of 80 HP the estimated maximum flow the pump station could deliver is approximately 990 GPM. Mainline sizes are sufficient to convey maximum flow.

Figure 3 illustrates the mainline routing and location used for the hydraulic estimation. Table 2 below illustrates the hydraulic analysis.

Table 2: Hydraulic Analysis

Hydraulic Calculation Worksheet

Aqua Engineering, Inc.
375 E. Horsetooth Rd
Fort Collins, CO 80525

By: JHK
Project Name: CU:NEWTON-ATHENS
Tap # or Hwy Segment: PUMP STATION

System Parameters:

Pump Discharge Pressure:	90	Max. System Flow:	800
Connection Elevation (Feet)	5304	Max RCV Flow	100
		Concurrent RCV's:	YES

COMPONENT FEATURES

	COMPONENT		Flow (GPM)	SIZE (ID) inch	QUANTITIY, LENGTH or HEIGHT(+/-)	H_L/unit	Calc. H_L (PSI)	
A.	Sprinkler				1	60	60	A.
B.	Lateral				1	5.00%	3	B.
C.	RCV		100	2"	1	3	3	C.
D.	Mainline Segment 1	4"	0	4.491	1	0.00 /100'	0.00	
		6"	0	5.348	1	0.00 /100'	0.00	
		8"	400	6.963	2000	0.22 /100'	4.45	
	Mainline Segment 2	4"	0	4.491	1	0.00 /100'	0.00	
		6"	0	5.348	1	0.00 /100'	0.00	
		8"	0	6.963	1	0.00 /100'	0.00	
					Subtotal D. Mainline Losses =		4.45	D.
E.	PUMP SYSTEM LOSSES		800	NA	1	5	5.00	E.
F.	Flow Sensor - NA		NA	NA	0	0.50 /100'	0.00	F.
G.	Water Meter - NA		NA	NA	1	0.00	0.00	G.
H.	Backflow - NA		NA	NA	1	0.00	0	H.
I.	Elevation @ Lateral (feet)				5321	0.433 PSI/FT	7.361	I.
J.	Service Pipe- NA		0	1.0	10.0	0.00	0.00	J.
K.	Misc Losses					5.00%	0.77	K.
					Total Pressure Required		83.58	
					Total Pressure Available		90	
					Excess/-Deficiet Pressure		6.42	

Opinions of Probable Cost

The opinions of probable costs are provided in Tables 3 and 4 below:

- Table 3 presents the cost for the irrigation pond and pump station.
- Table 4 presents the cost for the irrigation storage vessel and pump station.

Table 3: Opinion of Probable Cost – Irrigation Pond and Pump Station

NEWTON-ATHENS RAW WATER IRRIGATION SYSTEM MASTER PLAN
OPINION OF PROBABLE COST (IRRIGATION POND AND PUMP STATION)

Description	Units	Number	Unit Cost	Total Cost
IRRIGATION POND				
Excavation, Hauling, Disposal	CY	2,805	$8.00	$22,440.00
Lake Liner (Includes Grading, Liner, Geotextile)	SF	17,000	$4.50	$76,500.00
8" Supply Pipe	LF	350	$35.00	$12,250.00
12" Overflow Pipe (OPTION #1)	LF	300	$50.00	$15,000.00
12" Overflow Pipe (OPTION #2)	LF	400	$65.00	$26,000.00
Intake Structure	LS	1	$10,000.00	$10,000.00
16" Intake Pipe	LF	75	$80.00	$6,000.00
Wet Well	LS	1	$25,000.00	$25,000.00
Design Fees (10% of Construction Cost)	LS	1	$18,000.00	$18,000.00
SUBTOTAL 1 (Includes 20% Contingency and Overflow Pipe Option #1)				**$222,000.00**
SUBTOTAL 2 (Includes 20% Contingency and Overflow Pipe Option #2)				**$235,000.00**
IRRIGATION PUMP STATION				
Pump Station	EA	1	$125,000.00	$125,000.00
Pump Station Enclosure	EA	1	$70,000.00	$70,000.00
12-inch Mainline Pipe	LF	170	$50.00	$8,500.00
8-inch Mainline Pipe	LF	300	$40.00	$12,000.00
12-inch Isolation Gate Valve	EA	1	$9,000.00	$9,000.00
8-inch Isolation Gate Valve	EA	1	$6,000.00	$6,000.00
Connection to Existing Mainline Pipe	EA	1	$2,500.00	$2,500.00
Mainline Pipe Stub Out (Includes Winterization Assembly)	EA	1	$2,500.00	$2,500.00
Aeration System	LS	1	$15,000.00	$15,000.00
Electrical	LS	1	$30,000.00	$30,000.00
Design Fees (10% of Construction Cost)	LS	1	$28,000.00	$28,000.00
SUBTOTAL 3 (Includes 20% Contingency)				**$370,000.00**
TOTAL (SUBTOTAL 1 + SUBTOTAL 3)				**$592,000.00**
TOTAL (SUBTOTAL 2 + SUBTOTAL 3)				**$605,000.00**

NOTE:
1. This Opinion of Probable Construction Cost is not intended for use in bidding or ordering of equipment. Aqua Engineering will not be responsible for differences between this information and actual project equipment quantities or design/construction costs.

Table 4: Opinion of Probable Cost – Irrigation Storage Vessel and Pump Station

NEWTON-ATHENS RAW WATER IRRIGATION SYSTEM MASTER PLAN
OPINION OF PROBABLE COST (IRRIGATION STORAGE VESSEL AND PUMP STATION)

Description	Units	Number	Unit Cost	Total Cost
IRRIGATION STORAGE VESSEL				
Excavation, Hauling, Disposal	CY	9,259	$8.00	$74,074.07
5-Foot Dia. Corrugated HDPE and Manifold	LF	4,350	$300.00	$1,305,000.00
Grading and Backfill	CY	6,238	$3.00	$18,715.28
8" Supply Pipe	LF	350	$30.00	$10,500.00
12" Overflow Pipe (OPTION #1)	LF	300	$50.00	$15,000.00
12" Overflow Pipe (OPTION #2)	LF	400	$65.00	$26,000.00
Intake Structure or Filtration	LS	1	$30,000.00	$30,000.00
16" Intake Pipe	LF	75	$70.00	$5,250.00
Wet Well	LS	1	$30,000.00	$30,000.00
Design Fees (10% of Construction Cost)	LS	1	$150,000.00	$150,000.00
SUBTOTAL 1 (Includes 20% Contingency and Overflow Pipe Option #1)				**$1,966,000.00**
SUBTOTAL 2 (Includes 20% Contingency and Overflow Pipe Option #2)				**$1,979,000.00**
IRRIGATION PUMP STATION				
Pump Station	EA	1	$125,000.00	$125,000.00
Pump Station Enclosure	EA	1	$70,000.00	$70,000.00
12-inch Mainline Pipe	LF	170	$50.00	$8,500.00
8-inch Mainline Pipe	LF	300	$40.00	$12,000.00
12-inch Isolation Gate Valve	EA	1	$10,000.00	$10,000.00
8-inch Isolation Gate Valve	EA	1	$6,000.00	$6,000.00
Connection to Existing Mainline Pipe	EA	1	$2,500.00	$2,500.00
Mainline Pipe Stub Out (Includes Winterization Assembly)	EA	1	$2,500.00	$2,500.00
Electrical	LS	1	$30,000.00	$30,000.00
Design Fees (10% of Construction Cost)	LS	1	$27,000.00	$27,000.00
SUBTOTAL 3 (Includes 20% Contingency)				**$352,000.00**
TOTAL (SUBTOTAL 1 + SUBTOTAL 3)				**$2,318,000.00**
TOTAL (SUBTOTAL 2 + SUBTOTAL 3)				**$2,331,000.00**

NOTE:
1. This Opinion of Probable Construction Cost is not intended for use in bidding or ordering of equipment. Aqua Engineering will not be responsible for differences between this information and actual project equipment quantities or design/construction costs.

ARAPAHOE AVE.

CONNECT TO EXISTING GOSSAMER DITCH
PIPE UNDERNEATH ARAPAHOE AVENUE. INVERT UNKNOWN.

OVERFLOW PIPE (OPTION #2)

CAP MAINLINE FOR FUTURE USE FOR
WEST SIDE OF SITE

PROPOSED CONNECTION:
APPROXIMATE LOCATION OF EXISTING
CONCRETE VAULT, TYP.

APPROXIMATE SUPPLY PIPE
INVERT EL. 5307'

ROUTE AND CONNECT TO
DISCHARGE PIPING FROM
EXISTING PUMP HOUSE

CAP/PLUG EXISTING
12" CONCRETE PIPE

PROPOSED POND LOCATION
*WATER SURFACE EL. ±5304'

APPROXIMATE LOCATION
OF GOSSAMER DITCH (PIPE), TYP.

EXISTING PUMP HOUSE - REMOVE

APPROXIMATE OVERFLOW PIPE
INVERT EL. 5302'

OVERFLOW PIPE (OPTION #1)

CONNECT TO FOOTBALL FIELD
IRRIGATION MAINLINE PIPE

CAP/PLUG EXISTING
12" CONCRETE PIPE

FOOTBALL PRACTICE FIELDS

LEGEND

- EXISTING CONCRETE VAULT
- EXISTING RAW WATER DELIVERY PIPE
- SUPPLY PIPE *HDPE DR32.5 PE4710 *SIZE: PER PLAN
- INTAKE PIPE *HDPE DR32.5 PE4710 *SIZE: PER PLAN
- OVERFLOW PIPE *HDPE DR32.5 PE4710 *SIZE: PER PLAN
- MAINLINE PIPE *HDPE DR11 PE4710 *SIZE: PER PLAN
- INTAKE STRUCTURE
- PUMP STATION
- ISOLATION GATE VALVE ASSEMBLY
- WINTERIZATION ASSEMBLY
- CAPPED PIPE
- PIPE CONNECTION

100' 0' 50' 100' 200'

1" = 100'

SCALE MUST MEASURE 1 INCH OR DRAWING IS NOT TO SCALE.

Aqua Engineering Inc.
Innovative Water Solutions

University of Colorado
Newton and Athens Court Raw Water Master Plan
Boulder, CO

IRRIGATION POND AND PUMP CONCEPT

DATE	SEPTEMBER 4, 2015
DESIGNED BY	CBK/JHK
DRAWN BY	JHK
CHECKED BY	RWB

REVISIONS

SHEET NO.
FIGURE 1

File: C:\CU\SMITH\DGSS--IMP\NEWTON--ATHENS\P--Newton&AthensFIGURES.dwg printed by Cullen Kinoshita on 9/4/2015 11:16:38 AM using AutoCAD Version: 20.0s (LMS Tech)

ARAPAHOE AVE.

CONNECT TO EXISTING GOSSAMER DITCH
PIPE UNDERNEATH ARAPAHOE AVENUE. INVERT UNKNOWN.

OVERFLOW PIPE (OPTION #2)

PROPOSED CONNECTION:
APPROXIMATE LOCATION OF EXISTING
CONCRETE VAULT, TYP.

APPROXIMATE SUPPLY PIPE
INVERT EL. 5307'

CAP/PLUG EXISTING
12" CONCRETE PIPE

CAP MAINLINE FOR FUTURE USE FOR
WEST SIDE OF SITE

ROUTE AND CONNECT TO
DISCHARGE PIPING FROM
EXISTING PUMP HOUSE

EXISTING PUMP HOUSE - REMOVE

APPROXIMATE OVERFLOW PIPE
INVERT EL. 5302'

EXAMPLE VESSEL LOCATION
*GRADE EL. ±5304'
* TOP OF TANKS EL. ±5300'
* BOTTOM OF TANKS EL. ±5295'

OVERFLOW PIPE (OPTION #1)

CONNECT TO FOOTBALL FIELD
IRRIGATION MAINLINE PIPE

CAP/PLUG EXISTING
12" CONCRETE PIPE

FOOTBALL PRACTICE FIELDS

LEGEND

- EXISTING CONCRETE VAULT
- EXISTING RAW WATER DELIVERY PIPE
- SUPPLY PIPE
  *CORRUGATED HDPE
  *SIZE: PER PLAN
- INTAKE PIPE
  *CORRUGATED HDPE
  *SIZE: PER PLAN
- OVERFLOW PIPE
  *CORRUGATED HDPE
  *SIZE: PER PLAN
- MAINLINE PIPE
  *HDPE DR11 PE4710
  *SIZE: PER PLAN
- PUMP STATION
- ISOLATION GATE VALVE ASSEMBLY
- WINTERIZATION ASSEMBLY
- CAPPED PIPE
- PIPE CONNECTION

100' 0' 50' 100' 200'

1" = 100'

SCALE MUST MEASURE 1 INCH OR DRAWING IS NOT TO SCALE.

Aqua Engineering Inc.
Innovative Water Solutions

University of Colorado
Newton and Athens Court Raw Water Master Plan
Boulder, CO

IRRIGATION VESSEL AND PUMP CONCEPT

DATE	SEPTEMBER 4, 2015
DESIGNED BY	CBK/JHK
DRAWN BY	JHK
CHECKED BY	RWB

REVISIONS
-
-
-
-

SHEET NO.
FIGURE 2

Files: G:\CU\SMITH\GDSS--MP\NEWTON--ATHENS\P--Newton\Athens\FIGURES.dwg, printed by Cullen Kincaid\ta on 9/4/2015 11:16:53 AM, using AutoCAD Version: 20.0s (LMS Tech)

ARAPAHOE AVE.

UNIVERSITY AVE.

CONNECTION (FROM 12" STUB OUT)
EL. ±5304

ISOLATION GATE VALVE, TYP.

SUBMAINLINE CONNECTION
*INCLUDE ISOLATION GATE (NOT SHOWN)

AIR VACUUM RELIEF VALVE

HIGHEST ELEVATION (WORST CASE)
EL. ±5321

CLASS 200 PVC SUBMAINLINE (EXAMPLE)

APPROXIMATE MAINLINE ROUTING
*MAINLINE ASSUMED AS HDPE DR11 PE 4710

FOOTBALL PRACTICE FIELDS

A 1
50,150sf
1.15ac

A 2
26,600sf
.61ac

A 3

B 4
49,000sf
1.13ac

B 3
27,000sf
.61ac

B 6
15,000sf
.34ac

8" 3" 4" 12" 12" 8" 3" 8" 3" 3" 8" 8" 8" 8" 8" 8" 8"

BOULDER CREEK

125' 0' 62' 125' 250'

1" = 125'

SCALE MUST MEASURE 1 INCH OR DRAWING IS NOT TO SCALE.

Aqua Engineering Inc.
Innovative Water Solutions

University of Colorado

Newton and Athens Court Raw Water Master Plan

Boulder, CO

HYDRAULIC CONCEPT

DATE	SEPTEMBER 4, 2015
DESIGNED BY	CBK/JHK
DRAWN BY	JHK
CHECKED BY	RWB

REVISIONS

SHEET NO.

FIGURE 3

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